

COURSE 3 - DATA ENGINEERING 2026

C3-06 - Start Here

Exact study order, first-use installation and one-heavy-tool learner-machine profile.

Artifact	C3-06-START
Status	2026 - LEARNER EDITION
Phase	C3-06 Streaming & Event-Driven Pipelines 13 lessons 34-46 focused hours
Prerequisite	C3-05 Gate PASS; C2B streaming/lakehouse concepts may be recalled but Kafka/Spark streaming implementation is taught here.
Current lane	Apache Kafka 4.3.1 Spark Structured Streaming 4.2.0 Delta Lake 4.4.0 Debezium 3.6.2.Final
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny event example -> guided Kafka/streaming action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> replay/state/production reasoning -> evidence. A running stream is not mastery if duplicates, late data, offsets, checkpoints or sink state are wrong.

What you will build

By the end of C3-06 you can design and operate replayable event pipelines: Kafka topics/groups/offsets, schema evolution and CDC boundaries, duplicate/late-event correctness, Structured Streaming state/checkpoints, lakehouse sinks and operational monitoring.

First-use setup - install only when the lesson reaches it

- Use the WSL2/Linux workspace from C3-00. Java 17+ is required by Kafka 4.3.1; this course standard accepts Java 21.
- Kafka first use (L02): follow the official Kafka 4.3.1 Quick Start. The supported local command lane is KRaft; do not install ZooKeeper for this course.
- Practice starter Python: `python3 -m venv .venv && source .venv/bin/activate && python -m pip install -r requirements.txt`.
- Structured Streaming first use (L10): install/verify `pyspark==4.2.0` only in the Spark lane, then `python -c "import pyspark; print(pyspark.__version__)"`.
- Delta first use (L12): follow the Delta 4.4.0/Spark 4.2 compatibility instructions for the selected runtime. Do not assume plain PySpark automatically includes Delta.
- Keep Airflow and other heavy local services closed. On an 8 GB laptop, run Kafka and Spark sequentially unless a small integration exercise explicitly needs both.

If live-tool setup is blocked, continue the book/simulator/design work and mark that exact live evidence PENDING. Do not convert simulator output into Kafka/Spark/Delta proof.

Exact study route

- Open the C3-070 lesson at the exact Mentor page.
- Follow once using starter files.
- Attempt matching C3-071 Pxx and save genuine evidence.
- Only then open C3-072 Rxx; close it and complete the changed retry.
- After all 13 lessons, attempt C3-073 ShipmentStream Mini-Lab.
- Then attempt C3-074A MeterStream Gate A.
- If repair is needed, use targeted remediation + changed micro-retest. Gate B/C stay sealed until explicit full-retest assignment.

Laptop profile

STREAM profile: WSL2 + editor/browser + one heavy runtime at a time. Kafka 4.3.1 supports a simple single-broker KRaft learning setup, but that does not demonstrate production replication/availability. Spark Structured Streaming and Delta evidence is collected in a separate run when needed.